Policy gradient methods

Lecture 1: from gradient ascent to REINFORCE

Carlos Cotrini

ETH Zurich

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Part 1

Gradient ascent

Adds. The recap that opens the morning: a grasshopper climbs a hill by measuring the gradient and jumping uphill.

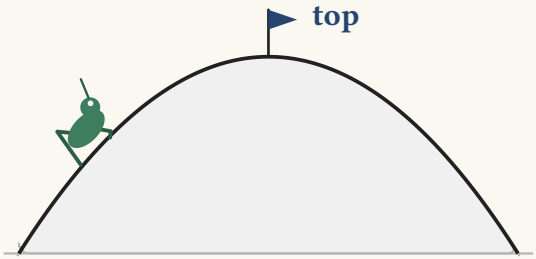
Answers. How a program finds the top of a hill when it can only feel the ground under its feet.

A grasshopper wants to reach the top of the hill



1 The grasshopper stands on the slope.

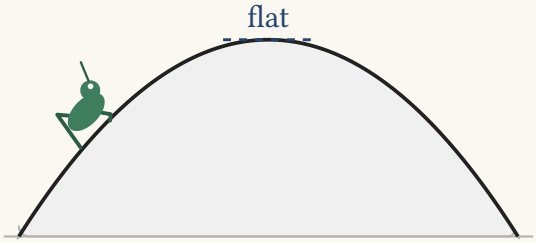
A grasshopper wants to reach the top of the hill



1 The grasshopper stands on the slope.

2 The top is the point of highest altitude: the place it wants to reach.

Keep jumping uphill until the ground is flat

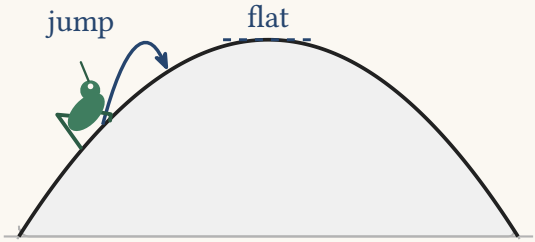


Idea

```
while not flat:
```

1 At the top the ground is flat: the grasshopper stops there.

Keep jumping uphill until the ground is flat



Idea

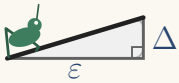
```
while not flat:
    jump in the direction
    that increases height
```

1 At the top the ground is flat: the grasshopper stops there.

2 Everywhere else it jumps uphill.

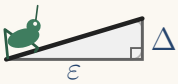
3 To know which way is uphill, it needs the gradient.

The gradient is the change in altitude per step



1 ε (epsilon) is the small step we probe with. Δ (delta) is the altitude change over it.

The gradient is the change in altitude per step



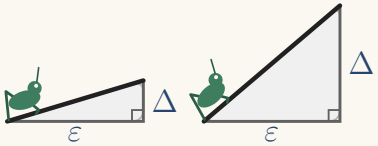
$$\text{Gradient} \approx \frac{\Delta}{\varepsilon} = \frac{\text{change in altitude}}{\text{step size}}$$

Here the step size is the probe step ε. The jump size α comes later.

$\Delta/\varepsilon \rightarrow$

1. ε (epsilon) is the small step we probe with. Δ (delta) is the altitude change over it.
2. The gradient is Δ/ε. We draw it as an arrow under each slope.

The gradient is the change in altitude per step



$$\text{Gradient} \approx \frac{\Delta}{\varepsilon} = \frac{\text{change in altitude}}{\text{step size}}$$

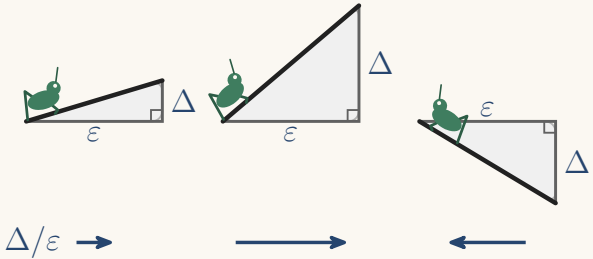
Here the step size is the probe step ε. The jump size α comes later.

Δ/ε →

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The gradient is the change in altitude per step

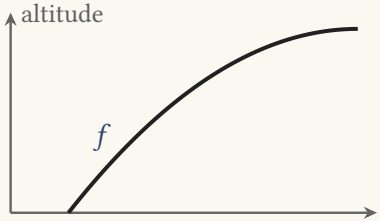


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Here the step size is the probe step ε. The jump size α comes later.

1. ε (epsilon) is the small step we probe with. Δ (delta) is the altitude change over it.
2. The gradient is Δ/ε. We draw it as an arrow under each slope.
3. Steeper means a longer arrow. When the altitude drops, the arrow points left.

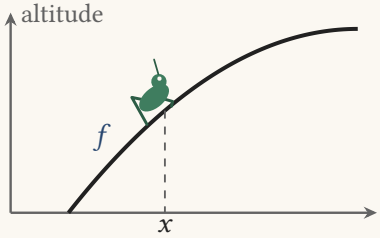
Python measures the gradient with a small step



function

```
def grad(f, x):
```

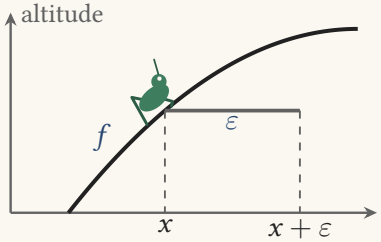
Python measures the gradient with a small step



function *location*

```
def grad(f, x):
```

Python measures the gradient with a small step



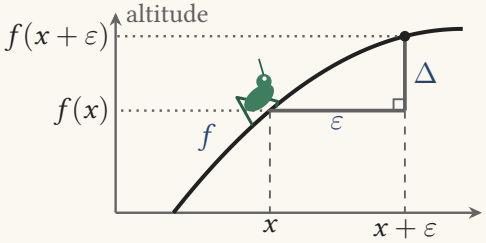
function *location*

```
def grad(f, x):
    eps = 0.01
```

small

1 `eps` is epsilon, the small step we probe with. The picture draws it large.

Python measures the gradient with a small step



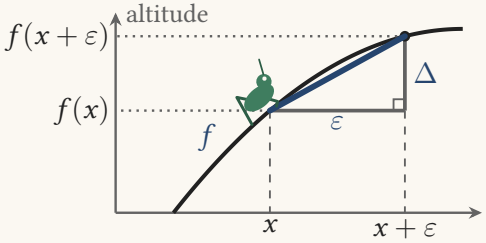
function *location*

```
def grad(f, x):
    eps = 0.01          ← small
    delta = f(x + eps) - f(x)
```

1 `eps` is epsilon, the small step we probe with. The picture draws it large.

2 `delta` is the change in altitude over that step: `f(x + eps)` minus `f(x)`.

Python measures the gradient with a small step



function *location*

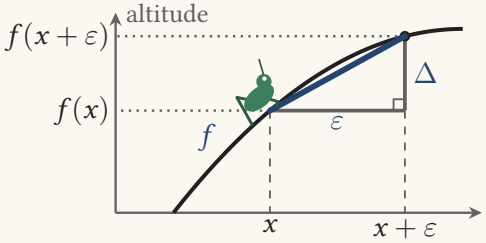
```
def grad(f, x):
    eps = 0.01          # small
    delta = f(x + eps) - f(x)
    grad = delta / eps
```

1 `eps` is epsilon, the small step we probe with. The picture draws it large.

2 `delta` is the change in altitude over that step: `f(x + eps)` minus `f(x)`.

3 `grad` is `delta / eps`, the slope of the segment between the two points.

Python measures the gradient with a small step



function *location*

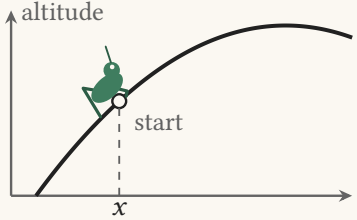
```
def grad(f, x):
    eps = 0.01          # small
    delta = f(x + eps) - f(x)
    grad = delta / eps
    return grad
```

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2 `delta` is the change in altitude over that step: `f(x + eps)` minus `f(x)`.

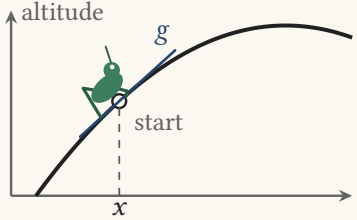
3 `grad` is `delta / eps`, the slope of the segment between the two points.

Gradient ascent measures, jumps and repeats



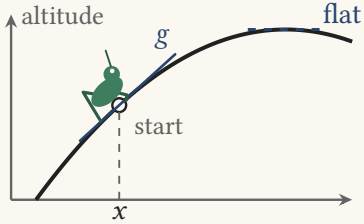

```
def gradient_ascent(f, x, alpha=0.1):
```

Gradient ascent measures, jumps and repeats




```
def gradient_ascent(f, x, alpha=0.1):
    g = grad(f, x)
```

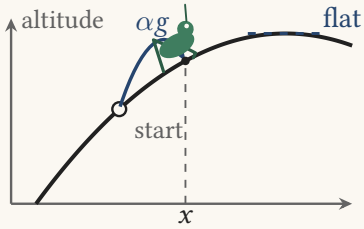
Gradient ascent measures, jumps and repeats




```
def gradient_ascent(f, x, alpha=0.1):
    g = grad(f, x)
    while abs(g) > 0.01:
```

1 g is the gradient at x. Flat means g is close to zero, here `abs(g)` at most 0.01.

Gradient ascent measures, jumps and repeats



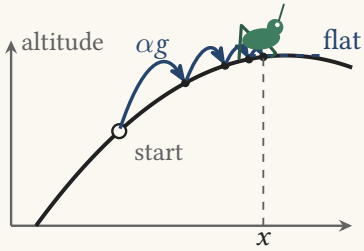

```
def gradient_ascent(f, x, alpha=0.1):
    g = grad(f, x)
    while abs(g) > 0.01:
        x = x + alpha * g
```

1 `g` is the gradient at `x`. Flat means `g` is close to zero, here `abs(g)` at most 0.01.

2 `alpha` is how far we jump per unit of gradient: the step size of the climb.

3 `eps` probes the ground, `alpha` sets the jump: they are different numbers.

Gradient ascent measures, jumps and repeats



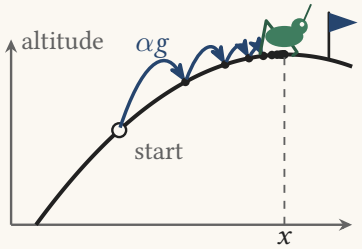

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2

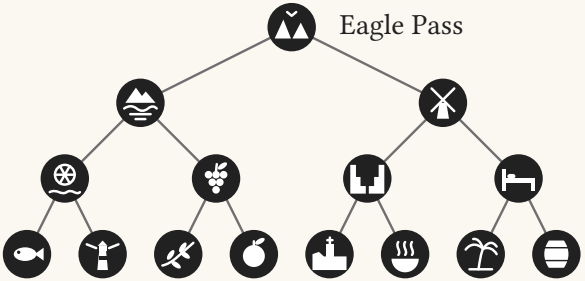
Part 2

Mara’s road

Adds. A first method that learns which road to take: a table of chances, turned into knobs and a softmax, climbed like a hill.

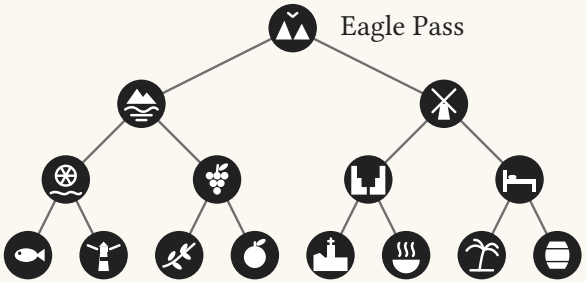
Answers. Playing every path several times works on a small town and stops working when a day has more places.

A town of fifteen places joined by roads



1 Fifteen places, joined by roads. A day follows one path down to the coast.

A town of fifteen places joined by roads



1 Fifteen places, joined by roads. A day follows one path down to the coast.

2 Play it yourself: scan the code or type the address.

`https://eth-cscs-hs26.github.io/CSCS-PAGE/viz/part2/courier-road/`

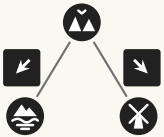
Mara plays her fiddle on the way down to the sea



1 Mara is a street fiddler. In one day she walks from Eagle Pass down to the sea, through three places.

Mara plays her fiddle on the way down to the sea



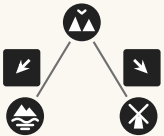


1 Mara is a street fiddler. In one day she walks from Eagle Pass down to the sea, through three places.

2 At every place the road forks: ↙ southwest or ↘ southeast. She picks one and walks on.

Mara plays her fiddle on the way down to the sea





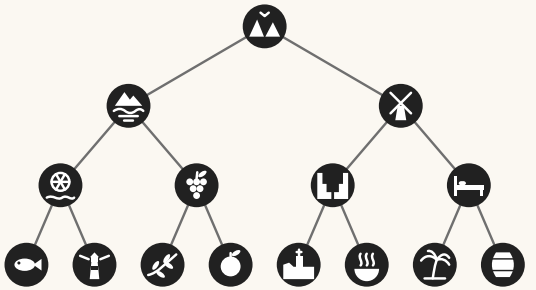


1 Mara is a street fiddler. In one day she walks from Eagle Pass down to the sea, through three places.

2 At every place the road forks: ↙ southwest or ↘ southeast. She picks one and walks on.

3 In every place she plays and the crowd drops coins in her hat. A place pays its usual coins, give or take one coin, whoever passes by. She wants as many coins as possible, but cannot know what a place will give until she has played there.

A simple strategy: play each path several times

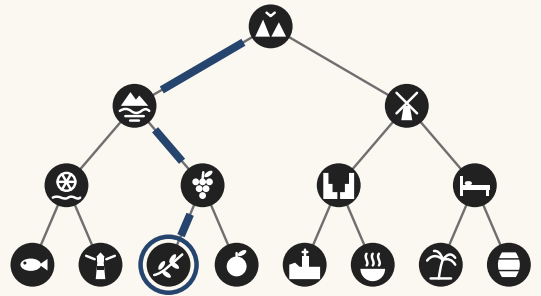


18.6 **18.8** **19.6** **19.2** **15.0** **19.4** **14.6** **14.4**

average coins of 5 days on each path

1 Luck makes the days differ. Play 5 days on each of the 8 paths and note the average of each.

A simple strategy: play each path several times

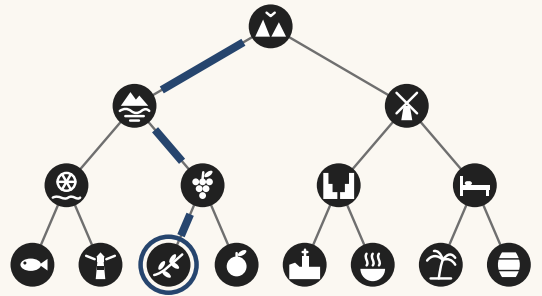


average coins of 5 days on each path

1 Luck makes the days differ. Play 5 days on each of the 8 paths and note the average of each.

2 Keep the best average. It finds the best path in 98% of the tries; one day per path only in 69%.

A simple strategy: play each path several times



average coins of 5 days on each path

1 Luck makes the days differ. Play 5 days on each of the 8 paths and note the average of each.

2 Keep the best average. It finds the best path in 98% of the tries; one day per path only in 69%.

3 Trying every path works here, since there are only 8. One more place on a path doubles that to 16, and each further place doubles it again.

Mara follows a table of chances



1 A table of chances: a row for each place where the road forks, a column for each road. Each row adds up to 1, and every cell starts at 0.5.

Mara follows a table of chances

	↙	↘
	0.5	0.5
	0.5	0.5
	0.5	0.5
	0.5	0.5
	0.5	0.5
	0.5	0.5
	0.5	0.5

1 A table of chances: a row for each place where the road forks, a column for each road. Each row adds up to 1, and every cell starts at 0.5.

2 At each place Mara takes each road with the chance in its cell. The table is her policy: her rule for choosing roads.

Mara follows a table of chances

	↙	↘
🏔	0.5	0.5
🌋	0.5	0.5
🌪	0.5	0.5
🎡	0.5	0.5
🍇	0.5	0.5
🏰	0.5	0.5
🛏	0.5	0.5

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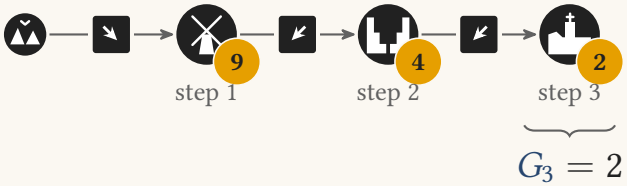
3 We use a method inspired by gradient ascent because it works well in practice on a wide variety of problems, and it generalises better than the simple strategy.

G_t is the sum of the coins from step t on



1 Gradient ascent needs an altitude. Ours is made of a day's coins, luck included.

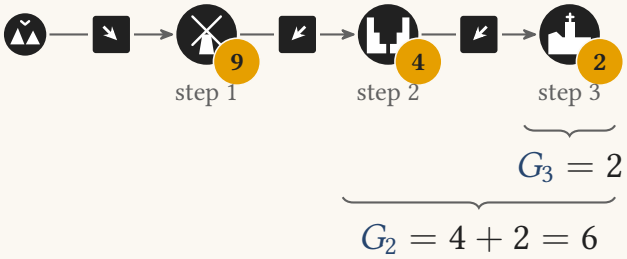
G_t is the sum of the coins from step t on



1 Gradient ascent needs an altitude. Ours is made of a day's coins, luck included.

2 G_t is the sum of the coins from step t to the end of the day.

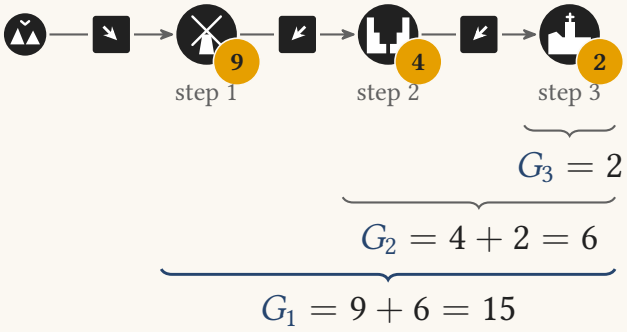
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1 Gradient ascent needs an altitude. Ours is made of a day's coins, luck included.

2 G_t is the sum of the coins from step t to the end of the day.

3 G_1 is the whole day: here 15 coins.

The same table gives different days

	step 1	step 2	step 3	G_1
day 1	9	4	2	15
day 2	7	7	3	17
day 3	8	5	1	14
day 4	9	3	3	15
day 5	10	3	6	19

1 One row is one day of the same table: its places and their coins.

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1 One row is one day of the same table: its places and their coins.

2 What changes: the roads, decided by the table's chances, and the luck of each place.

The same table gives different days

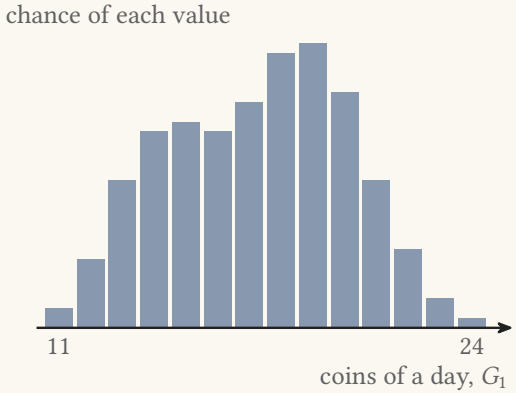
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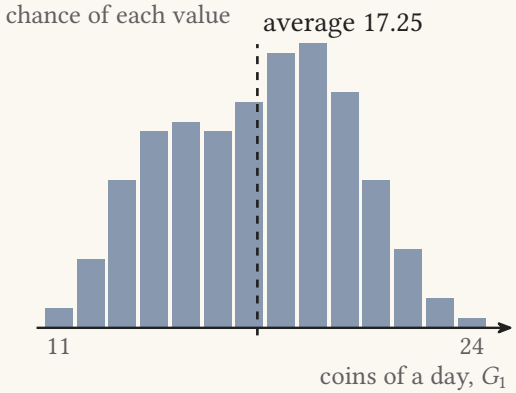
3 So G_1 is random: a different number every day.

A random number cannot be maximised



1 On many days the untouched table gives G_1 anywhere from 11 to 24 coins, some values more often than others.

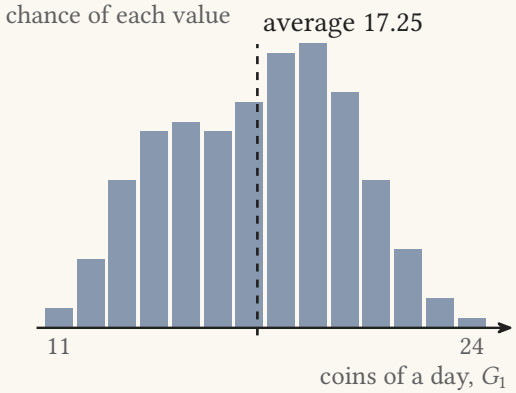
A random number cannot be maximised



1 On many days the untouched table gives G_1 anywhere from 11 to 24 coins, some values more often than others.

2 The average of G_1 over all the days it can give is 17.25 coins.

A random number cannot be maximised

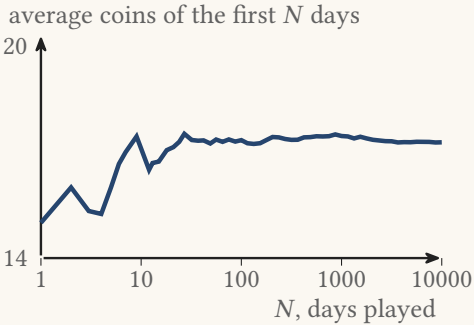


1 On many days the untouched table gives G_1 anywhere from 11 to 24 coins, some values more often than others.

2 The average of G_1 over all the days it can give is 17.25 coins.

3 A random number cannot be maximised: asking which table gives the biggest G has no answer.

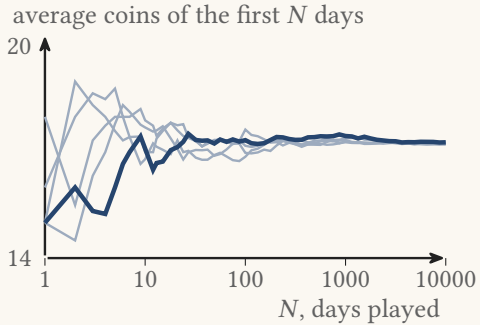
Our objective is the long-run average $\mathbb{E}[G_1]$



1 Let π_w be the policy of the table w. Play it on N days: the n-th day pays $G_1^{(n)}$.

Our objective is the long-run average $\mathbb{E}[G_1]$

$$\mathbb{E}[G_1] = \lim_{N\to\infty} \frac{1}{N} \sum_{n=1}^{N} G_1^{(n)}$$



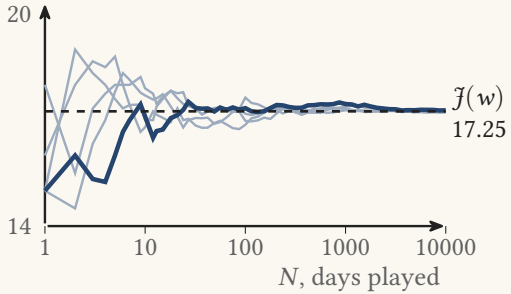
1 Let π_w be the policy of the table w. Play it on N days: the n-th day pays $G_1^{(n)}$.

2 The average of $G_1^{(1)}, \ldots, G_1^{(N)}$ settles as N grows, at a number we call $\mathbb{E}[G_1]$.

Our objective is the long-run average $\mathbb{E}[G_1]$

$$\mathbb{E}[G_1] = \lim_{N\to\infty} \frac{1}{N} \sum_{n=1}^{N} G_1^{(n)}$$

average coins of the first N days

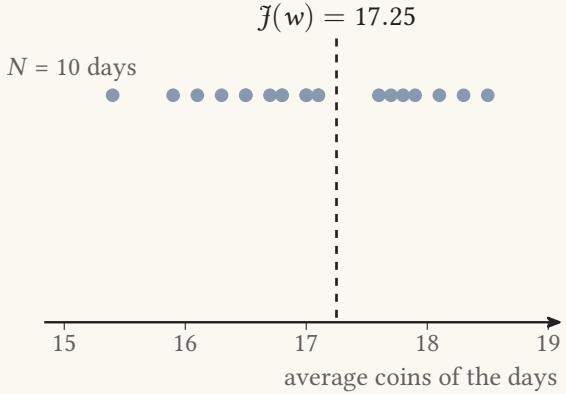


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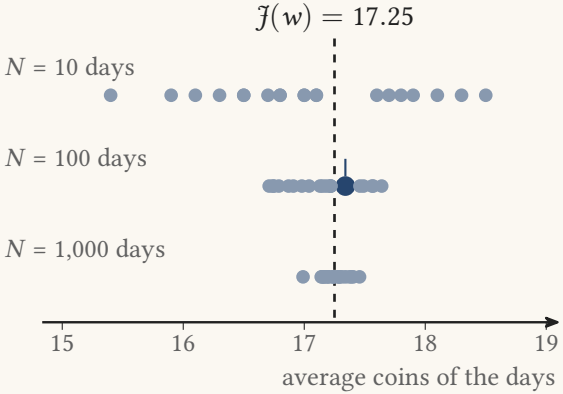
3 $\mathcal{J}(w) = \mathbb{E}[G_1]$ is the altitude of the grasshopper, and our objective is to maximise it.

More days bring the average closer to $\mathcal{J}(w)$



1 Each dot is the average of N days in one of 20 runs. The line is $\mathcal{J}(w)$.

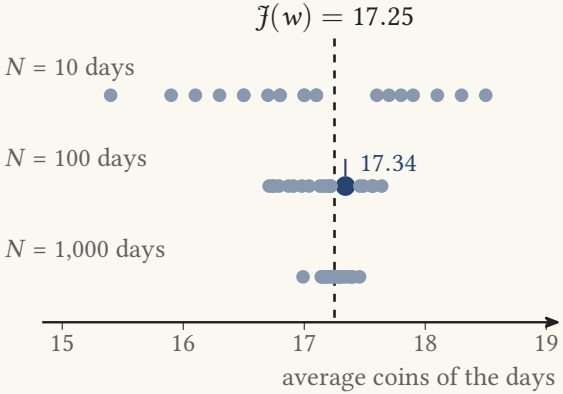
More days bring the average closer to $J(w)$



1 Each dot is the average of N days in one of 20 runs. The line is $J(w)$.

2 The dots miss the line by about 0.89 coins at N = 10, 0.28 at N = 100 and 0.09 at N = 1,000: the error shrinks as N grows.

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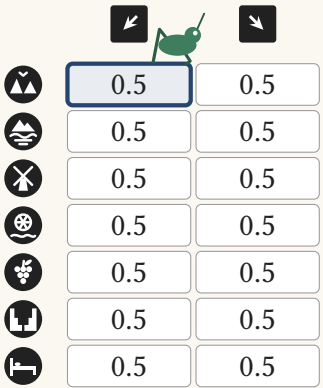
3 The first run of 100 days averages 17.34: our estimate of $J(w)$ from now on.

We pick one cell of the table at random

	↙	↘
	0.5	0.5
	0.5	0.5
	0.5	0.5
	0.5	0.5
	0.5	0.5
	0.5	0.5
	0.5	0.5

1 Pick one cell of the table at random.

We pick one cell of the table at random



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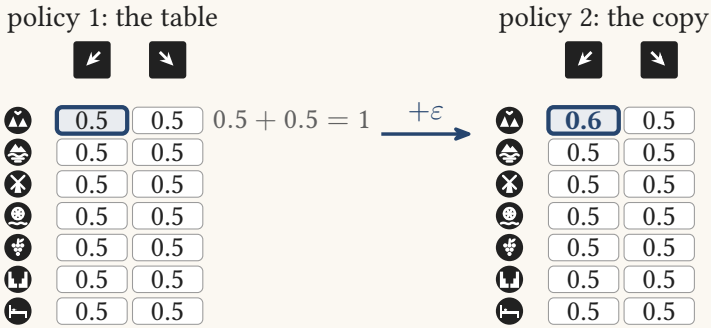
2 The chosen cell plays the grasshopper's part: its number is the spot where the grasshopper stands.

A copy of the table has ε added to the cell

policy 1: the table

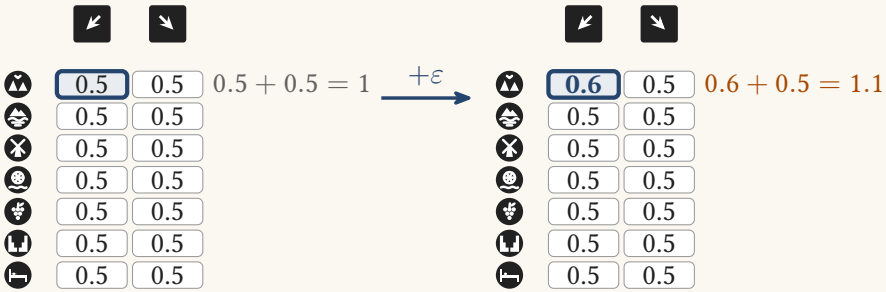
	↙	↘	
	0.5	0.5	$0.5 + 0.5 = 1$
	0.5	0.5	
	0.5	0.5	
	0.5	0.5	
	0.5	0.5	
	0.5	0.5	
	0.5	0.5	

A copy of the table has ε added to the cell



1 Policy 2 is the table with a small number $\varepsilon = 0.1$ added to the chosen cell.

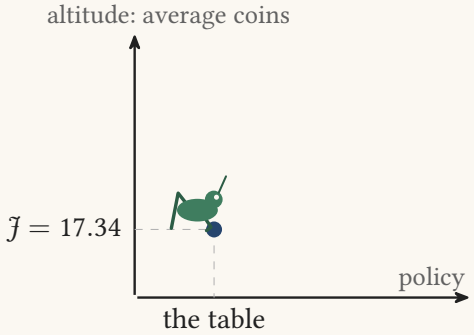
A copy of the table has ε added to the cell



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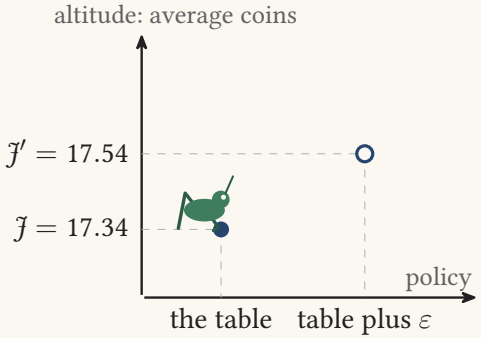
2 In the copy that row adds up to 1.1. We fix this in a moment.

Two averages over N days measure the gradient



1 Play $N = 100$ days with the table. Dice decide the roads and the luck of each place. Their average, $\mathcal{J} = 17.34$, estimates the altitude $\mathcal{J}(w)$.

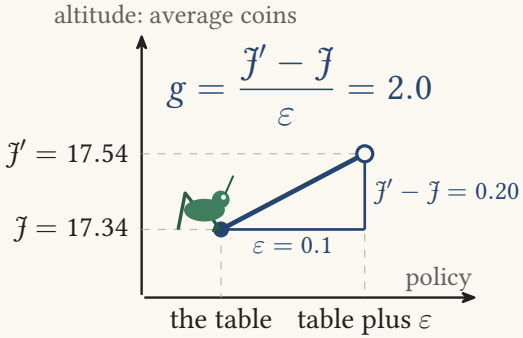
Two averages over N days measure the gradient



1 Play $N = 100$ days with the table. Dice decide the roads and the luck of each place. Their average, $\mathcal{J} = 17.34$, estimates the altitude $\mathcal{J}(w)$.

2 Play the same days again, with the same dice, on the table plus $\varepsilon = 0.1$ in the chosen cell: $\mathcal{J}' = 17.54$.

Two averages over N days measure the gradient



1 Play $N = 100$ days with the table. Dice decide the roads and the luck of each place. Their average, $\mathcal{J} = 17.34$, estimates the altitude $\mathcal{J}(w)$.

2 Play the same days again, with the same dice, on the table plus $\varepsilon = 0.1$ in the chosen cell: $\mathcal{J}' = 17.54$.

3 The same dice give both policies the same luck, so only the change shows. The measured gradient: $g = (\mathcal{J}' - \mathcal{J})/\varepsilon = 2.0$.

The cell jumps by α times the gradient

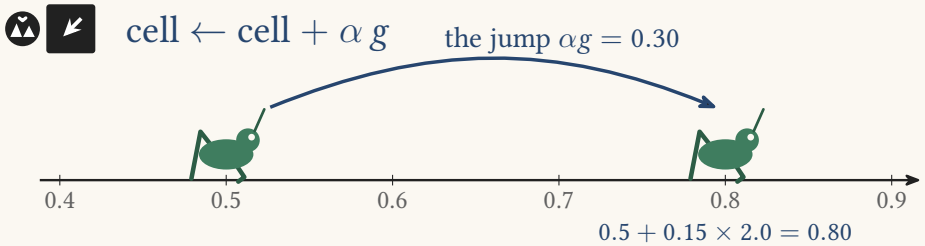


$$\text{cell} \leftarrow \text{cell} + \alpha\, g$$



1 If the copy earned more, g is positive and the cell goes up; otherwise it goes down.

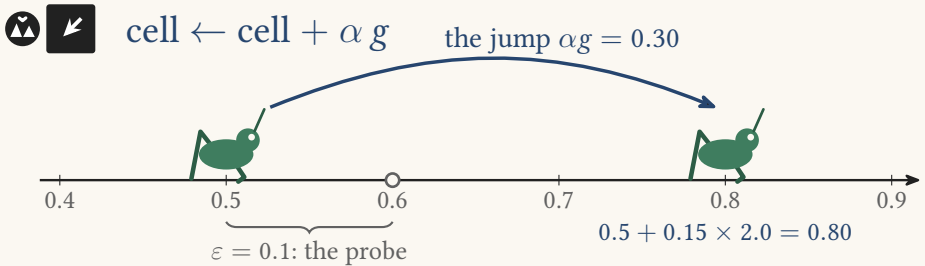
The cell jumps by α times the gradient



1 If the copy earned more, g is positive and the cell goes up; otherwise it goes down.

2 α is how far we jump per unit of gradient.

The cell jumps by α times the gradient



1 If the copy earned more, g is positive and the cell goes up; otherwise it goes down.

2 α is how far we jump per unit of gradient.

3 ε and α are two different small numbers.

The whole method is a short loop

```
repeat:
  pick a cell of the table at random
```

1 The chosen cell is the grasshopper.

The whole method is a short loop

```
repeat:
(1)   pick a cell of the table at random
      make a copy, add eps to that cell
(2)   play N days with each, same dice
      J, J' = average coins of each
```

1 The chosen cell is the grasshopper.

2 Two policies, and their two altitudes.

The whole method is a short loop

```
repeat:
  pick a cell of the table at random
  make a copy, add eps to that cell
  play N days with each, same dice
  J, J' = average coins of each
  g = (J' - J) / eps
  cell = cell + alpha * g
```

1 The chosen cell is the grasshopper.

2 Two policies, and their two altitudes.

3 The measured gradient, and the jump.

Knobs fix the problem: any number is allowed

chances

	↙	↘
	0.5	0.5
	0.5	0.5
	0.5	0.5
	0.5	0.5
	0.5	0.5
	0.5	0.5
	0.5	0.5

1 Adding ε to a chance can break a row: it must add up to 1.

Knobs fix the problem: any number is allowed

knobs w

	↙	↘
	0	0
	0	0
	0	0
	0	0
	0	0
	0	0
	0	0

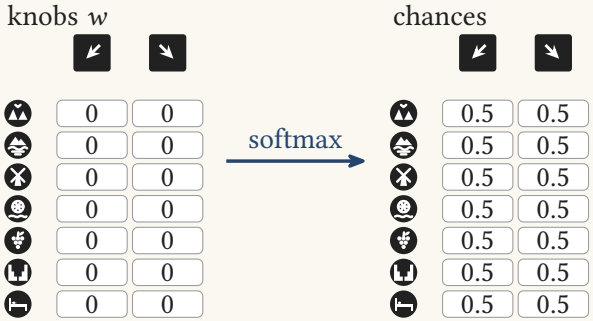
chances

	↙	↘
	0.5	0.5
	0.5	0.5
	0.5	0.5
	0.5	0.5
	0.5	0.5
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1 Adding ε to a chance can break a row: it must add up to 1.

2 So every cell holds a knob: any number. All knobs start at 0.

Knobs fix the problem: any number is allowed



1 Adding ε to a chance can break a row: it must add up to 1.

2 So every cell holds a knob: any number. All knobs start at 0.

3 Turn the knobs, then convert each row into chances: the softmax.

A bigger knob favours its road



knob
any number

↙	↘
3	−2

1 One row of the table: a knob for each road. A knob is any number, positive or negative.

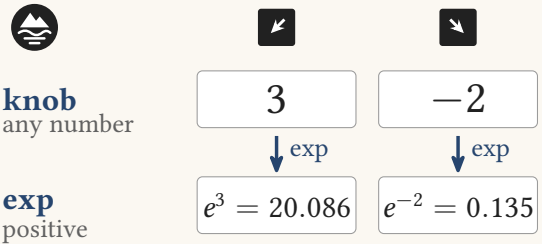
A bigger knob favours its road

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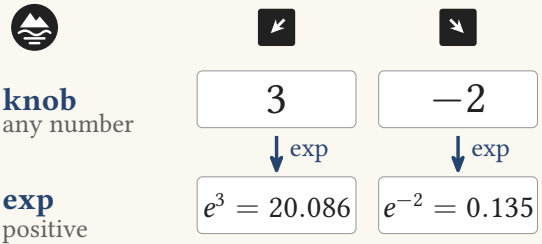
2 A bigger knob favours its road. Here ↙ is favoured. Two steps turn the knobs into chances.

Step 1: take the exponential of each knob



1 The exponential e^x turns any number into a positive one.

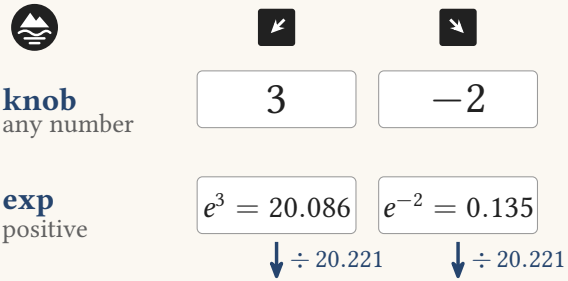
Step 1: take the exponential of each knob



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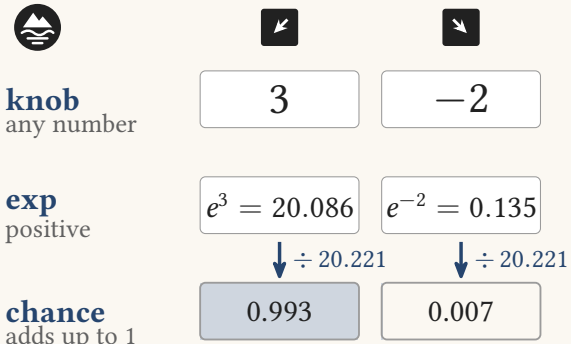
2 A large knob becomes a large number. A negative knob becomes a number close to 0.

Step 2: divide each by the sum to get chances



1 Add the two numbers, $20.086 + 0.135 = 20.221$, and divide each by this sum.

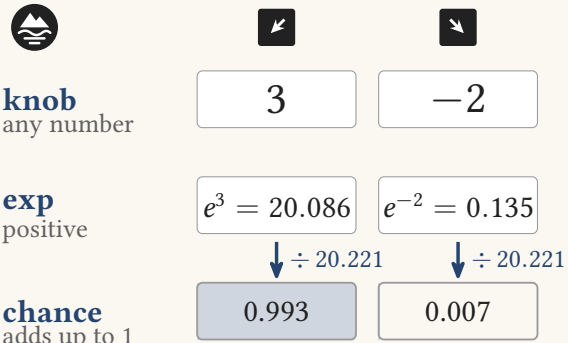
Step 2: divide each by the sum to get chances



1 Add the two numbers, $20.086 + 0.135 = 20.221$, and divide each by this sum.

2 The chances add up to 1. Knobs 3 and -2 make the ↙ road nearly certain.

Step 2: divide each by the sum to get chances



$$\text{chance}_i = \frac{e^{w_i}}{e^{w_1} + e^{w_2}}$$

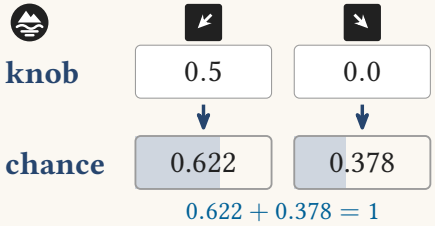
the softmax of a row of two knobs

1 Add the two numbers, $20.086 + 0.135 = 20.221$, and divide each by this sum.

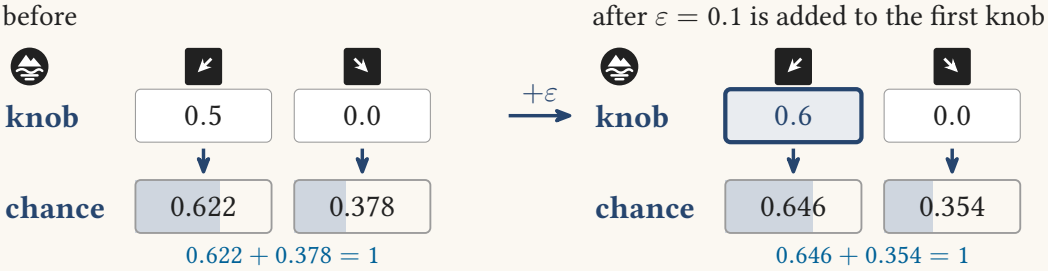
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Adding ε to a knob always leaves a valid row

before

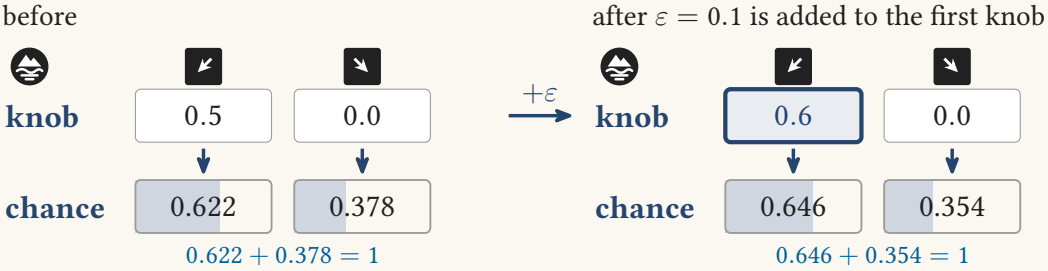


Adding ε to a knob always leaves a valid row



1 The first road becomes a little more likely, and the second a little less.

Adding ε to a knob always leaves a valid row



1 The first road becomes a little more likely, and the second a little less.

2 Whatever the knobs, a row adds up to 1 and each chance stays between 0 and 1.

The same loop turns the knobs

```
w = knobs, all 0
repeat:
   pick a cell of w at random
```

1 The table of knobs is where the grasshopper stands.

2 One pass of the loop is one hop of the grasshopper.

The same loop turns the knobs

```
w = knobs, all 0
repeat:
   pick a cell of w at random
   w2 = w, add eps to that cell
   play N days with softmax(w)
   play N days with softmax(w2)
   (the same dice for both)
   J, J' = average coins of each
   g = (J' - J) / eps
   w[cell] = w[cell] + alpha * g
```

1 The table of knobs is where the grasshopper stands.

2 One pass of the loop is one hop of the grasshopper.

3 The softmax turns every row of knobs into chances.

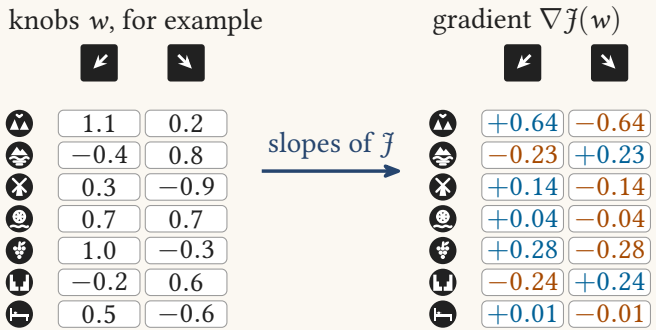
The table of all gradients is called $\nabla J(w)$

knobs w, for example

	↙	↘
	1.1	0.2
	−0.4	0.8
	0.3	−0.9
	0.7	0.7
	1.0	−0.3
	−0.2	0.6
	0.5	−0.6

1 These knobs w give $J(w) = 19.03$ coins a day on average.

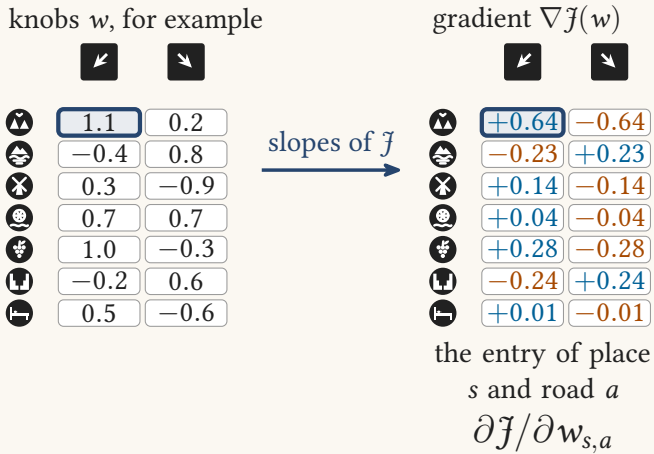
The table of all gradients is called $\nabla J(w)$



1 These knobs w give $J(w) = 19.03$ coins a day on average.

2 Each knob has a gradient, the slope of J in that knob. The table of all 14 is called $\nabla J(w)$.

The table of all gradients is called $\nabla \mathcal{J}(w)$



1 These knobs w give $\mathcal{J}(w) = 19.03$ coins a day on average.

2 Each knob has a gradient, the slope of $\mathcal{J}$ in that knob. The table of all 14 is called $\nabla \mathcal{J}(w)$.

3 The loop of the last frames measured one entry at a time.

A formula estimates the whole table from N days

$$\nabla J(w) \;\approx\; \frac{1}{N} \sum_{n=1}^{N} \sum_{t=1}^{T} \boxed{G_t^{(n)}} \, \nabla_w \log \pi_w(a_t^{(n)} \mid s_t^{(n)})$$

1 Play N days with π_w, each of $T = 3$ steps. Step t starts at place s_t and takes road a_t.

A formula estimates the whole table from N days

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the loop N N N N N N N N N N N N N N

the formula N

1 Play N days with π_w, each of $T = 3$ steps. Step t starts at place s_t and takes road a_t.

2 The loop plays N more days for every entry. The formula uses one set for all 14.

A formula estimates the whole table from N days

$$\nabla J(w) \approx \frac{1}{N} \sum_{n=1}^{N} \sum_{t=1}^{T} \boxed{G_t^{(n)}} \nabla_w \log \pi_w(a_t^{(n)} \mid s_t^{(n)})$$

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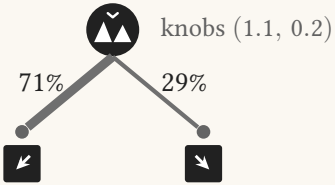
the formula N

1 Play N days with π_w, each of $T = 3$ steps. Step t starts at place s_t and takes road a_t.

2 The loop plays N more days for every entry. The formula uses one set for all 14.

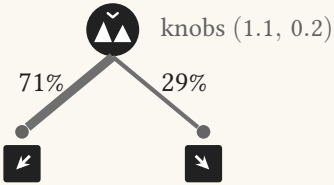
3 With N days it is an estimate. It settles as N grows, like the average coins did.

The score says how to turn the knobs



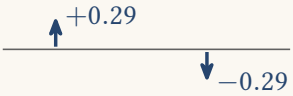
1 At the pass the knobs make the chances 71% and 29%.

The score says how to turn the knobs



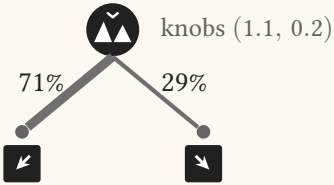
1 At the pass the knobs make the chances 71% and 29%.

2 Mara takes ↙ (71%). Its score says how to turn the knobs to make ↙ likelier.



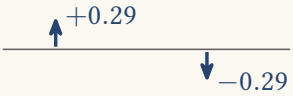
the other 12 knobs of the table get 0

The score says how to turn the knobs



1 At the pass the knobs make the chances 71% and 29%.

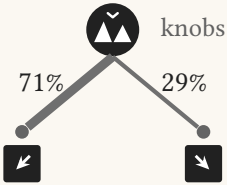
2 Mara takes ↙ (71%). Its score says how to turn the knobs to make ↙ likelier.



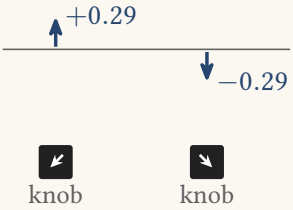
the other 12 knobs of the table get 0

3 The entry of road b is $\mathbf{1}[b = a_t] - \pi_w(b \mid s_t)$: the knob of the road taken goes up, the other knob goes down by as much.

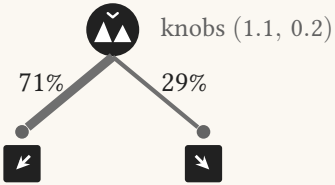
A rare road has a large score



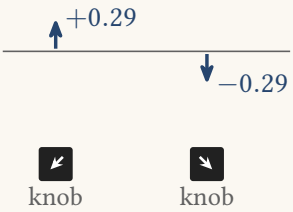
1 Mara takes ↙ (71%): a small score



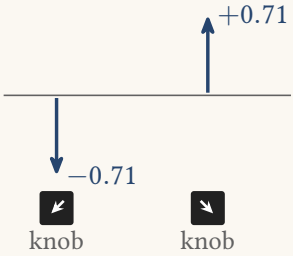
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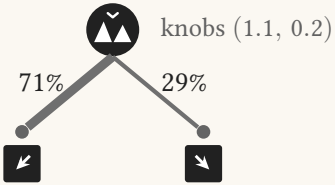
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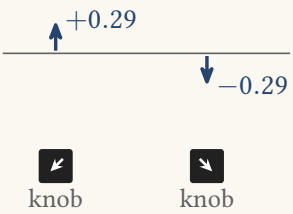
2 Mara takes ↘ (29%): a large score



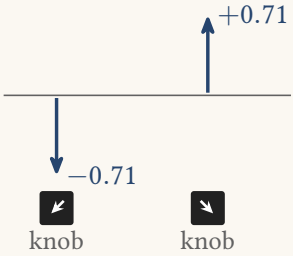
A rare road has a large score



1 Mara takes ↙ (71%): a small score



2 Mara takes ↘ (29%): a large score



3 The rarer the road taken, the larger the score: a rare road moves its knob far.

The gradient averages multiplier times score

$$\nabla \mathcal{J}(w) \approx$$

one entry for each place s and road a

1 $\nabla \mathcal{J}(w)$ says how much to adjust the knobs w to increase $\mathcal{J}$.

The gradient averages multiplier times score

$$\underbrace{\nabla \mathcal{J}(w)}_{\substack{\text{one entry for each} \\ \text{place } s \text{ and road } a}} \approx \overbrace{\frac{1}{N} \sum_{n=1}^{N}}^{\substack{\text{average over} \\ \text{the } N \text{ days}}}$$

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The gradient averages multiplier times score

$$\underbrace{\nabla \mathcal{J}(w)}_{\text{one entry for each place } s \text{ and road } a} \approx \overbrace{\frac{1}{N} \sum_{n=1}^{N}}^{\text{average over the } N \text{ days}} \underbrace{\sum_{t=1}^{T}}_{\text{sum over the steps}}$$

1 $\nabla \mathcal{J}(w)$ says how much to adjust the knobs w to increase $\mathcal{J}$.

The gradient averages multiplier times score

average over the N days

multiplier: the coins from step t on

$$\underbrace{\nabla \mathcal{J}(w)}_{\text{one entry for each place } s \text{ and road } a} \approx \frac{1}{N} \sum_{n=1}^{N} \underbrace{\sum_{t=1}^{T}}_{\text{sum over the steps}} \boxed{G_t^{(n)}}$$

1 $\nabla \mathcal{J}(w)$ says how much to adjust the knobs w to increase $\mathcal{J}$.

The gradient averages multiplier times score

average over the N days

multiplier: the coins from step t on

$$\nabla \mathcal{J}(w) \approx \frac{1}{N} \sum_{n=1}^{N} \sum_{t=1}^{T} G_t^{(n)} \nabla_w \log \pi_w(a_t^{(n)} \mid s_t^{(n)})$$

one entry for each place s and road a

sum over the steps

score of the road a_t taken at place s_t

1 $\nabla \mathcal{J}(w)$ says how much to adjust the knobs w to increase $\mathcal{J}$.

2 It is an average that grows with G_t and with the score of the road taken.

The multiplier G_t can be replaced by Ψ_t

$$\nabla J(w) \approx \frac{1}{N} \sum_{n=1}^{N} \sum_{t=1}^{T} \boxed{G_t^{(n)}} \nabla_w \log \pi_w(a_t^{(n)} \mid s_t^{(n)})$$

The multiplier G_t can be replaced by Ψ_t

$$\nabla \mathcal{J}(w) \approx \frac{1}{N} \sum_{n=1}^{N} \sum_{t=1}^{T} \boxed{G_t^{(n)}} \nabla_w \log \pi_w(a_t^{(n)} \mid s_t^{(n)})$$

replace G_t by Ψ_t

$$\nabla \mathcal{J}(w) \approx \frac{1}{N} \sum_{n=1}^{N} \sum_{t=1}^{T} \boxed{\Psi_t^{(n)}} \nabla_w \log \pi_w(a_t^{(n)} \mid s_t^{(n)})$$

1 It has been shown that the formula still works with another multiplier, written Ψ_t.

The multiplier G_t can be replaced by Ψ_t

$$\nabla \mathcal{J}(w) \approx \frac{1}{N} \sum_{n=1}^{N} \sum_{t=1}^{T} \boxed{G_t^{(n)}} \nabla_w \log \pi_w(a_t^{(n)} \mid s_t^{(n)})$$

↓ replace G_t by Ψ_t

$$\nabla \mathcal{J}(w) \approx \frac{1}{N} \sum_{n=1}^{N} \sum_{t=1}^{T} \boxed{\Psi_t^{(n)}} \nabla_w \log \pi_w(a_t^{(n)} \mid s_t^{(n)})$$

1 It has been shown that the formula still works with another multiplier, written Ψ_t.

2 There are several options for Ψ_t. The first is the one we have: $\Psi_t = G_t$.

3

Part 3

REINFORCE

Adds. The multiplier of the gradient formula is G_t, the coins from step t on: every day pushes every road it took as hard as the coins that came after it.

Answers. Part 2 played two batches of days to measure the slope of one knob; one batch of days can inform all of them.

$\Psi_t = G_t$ gives REINFORCE

$$\underbrace{w \leftarrow w + \alpha}_{1} \quad \frac{1}{N} \sum_{n=1}^{N} \sum_{t=1}^{T} \quad \boxed{\Psi_t} \quad \nabla_w \log \pi_w(a_t^{(n)} \mid s_t^{(n)})$$

1 Gradient ascent again: move the knobs by α times the gradient.

$\Psi_t = G_t$ gives REINFORCE

$$\underbrace{w \leftarrow w + \alpha}_{1} \quad \underbrace{\frac{1}{N}\sum_{n=1}^{N}\sum_{t=1}^{T}}_{2} \quad \boxed{\Psi_t} \quad \nabla_w \log \pi_w(a_t^{(n)} \mid s_t^{(n)})$$

1 Gradient ascent again: move the knobs by α times the gradient.

2 The average over N days and $T = 3$ steps is the estimate of the gradient of the last frames.

$\Psi_t = G_t$ gives REINFORCE

$$w \leftarrow w + \alpha \underbrace{\frac{1}{N}\sum_{n=1}^{N}\sum_{t=1}^{T}}_{2} \underbrace{\boxed{G_t^{(n)}}}_{3} \nabla_w \log \pi_w(a_t^{(n)} \mid s_t^{(n)})$$

1 Gradient ascent again: move the knobs by α times the gradient.

2 The average over N days and $T = 3$ steps is the estimate of the gradient of the last frames.

3 The multiplier's box holds G_t, the coins from step t to the end of the day: the first option of the last frame.

REINFORCE is a short loop

```
w = knobs, all 0
repeat:
  play N days with softmax(w)
```

1 *N* days with the current knobs. No copy of the table and no ε.

REINFORCE is a short loop

```
w = knobs, all 0
repeat:
  play N days with softmax(w)
  push every road of every day
    as hard as the coins after it
```

1 N days with the current knobs. No copy of the table and no ε.

2 Every road taken is pushed as hard as G_t, the coins that came after it.

REINFORCE is a short loop

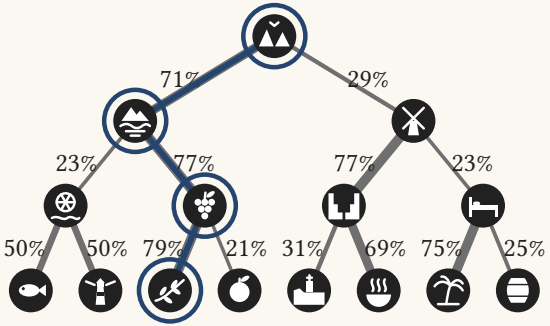
```
w = knobs, all 0
repeat:
1   play N days with softmax(w)
    push every road of every day
2     as hard as the coins after it
3   w = w + alpha * average push
```

1 N days with the current knobs. No copy of the table and no ε.

2 Every road taken is pushed as hard as G_t, the coins that came after it.

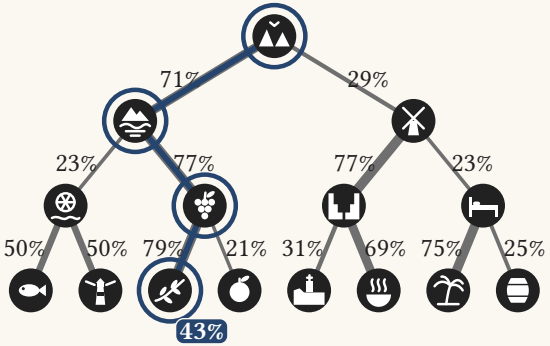
3 The knobs move by α times the average push.

Mara’s knobs make the 21-coin day likely



1 Each road is as thick as its chance. The numbers are the chances of the two roads of a place.

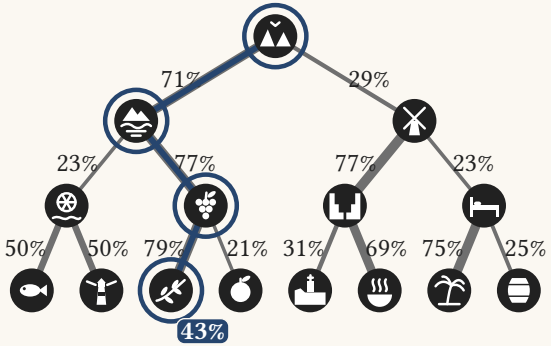
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2 These knobs make the best day likely: the path that pays 21 coins on average is taken on 43 days in 100.

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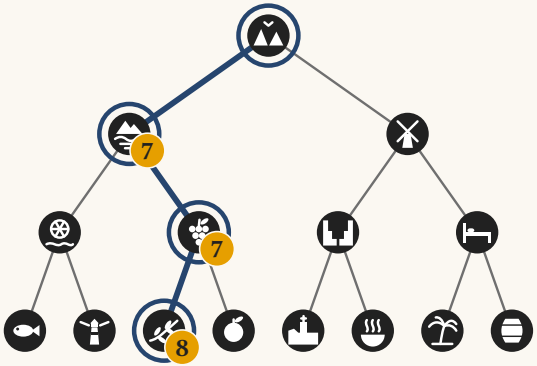


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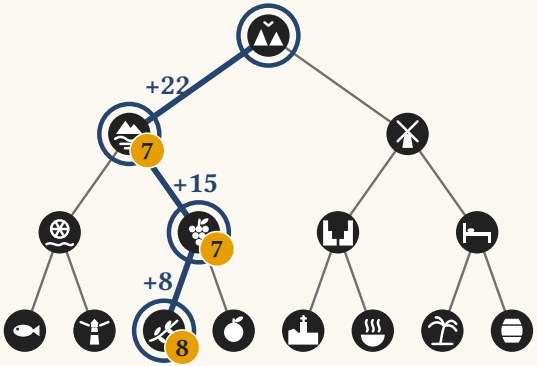
3 Their average day pays $\mathcal{J}(w) = 19.03$ coins. A few days will move the knobs.

Push each road by the coins that came after it



1 One day: → → . It was paid $7 + 7 + 8 = 22$ coins, luck included.

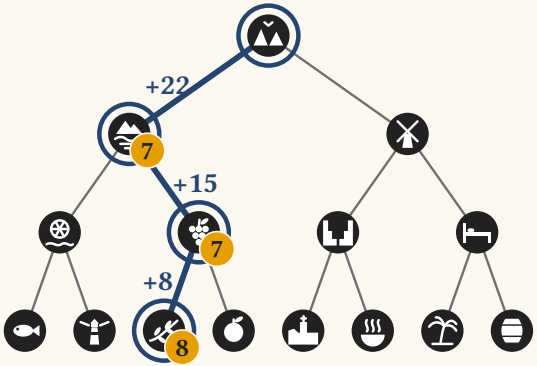
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Push each road by the coins that came after it

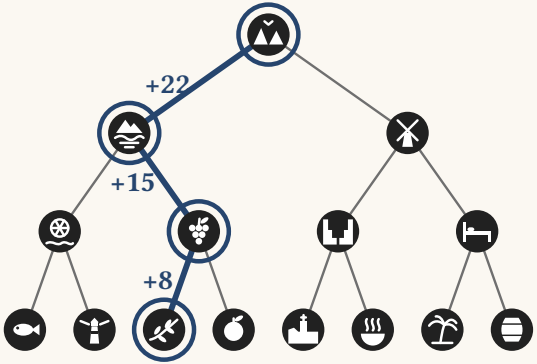


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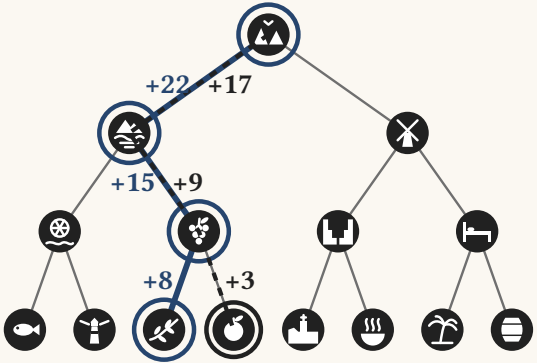
2 Each road is pushed up by the coins collected from it on: 22, then 15, then 8.

3 A road gets credit for what came after it. The coins collected before it do not count.

A richer day pushes the shared roads harder

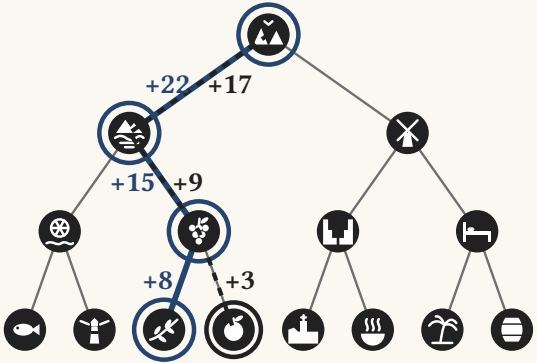


A richer day pushes the shared roads harder



1 A second day (dashed): ⛰ → 🍇 → 🍎. It was paid $8 + 6 + 3 = 17$ coins: pushes of 17, 9 and 3.

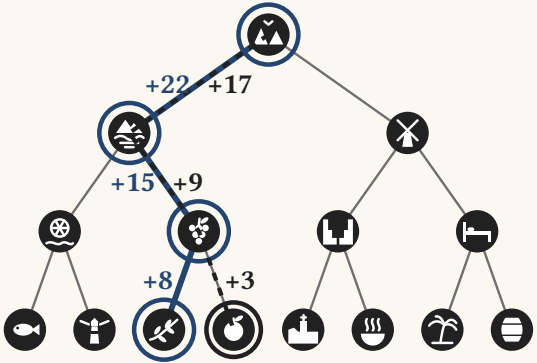
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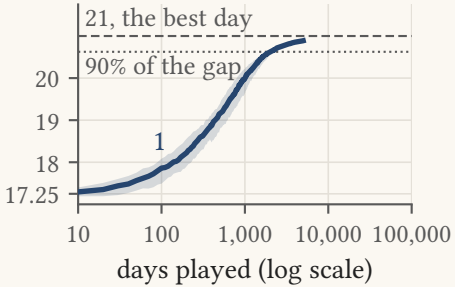


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2 The days share two roads. The richer day pushes them harder: 22 against 17, 15 against 9.

3 At they split: is pushed by 8, by 3.

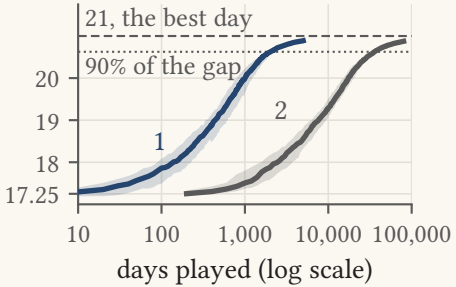
REINFORCE learns from days alone



Median of 200 runs and their middle half. $\alpha = 0.0316$ is the fastest step at which 99 of 100 other runs close 90% of the gap.

1 REINFORCE (10 days an update, $\alpha = 0.0316$) closes 90% of the gap from 17.25 to 21 coins in a median of 1,950 days.

REINFORCE learns from days alone

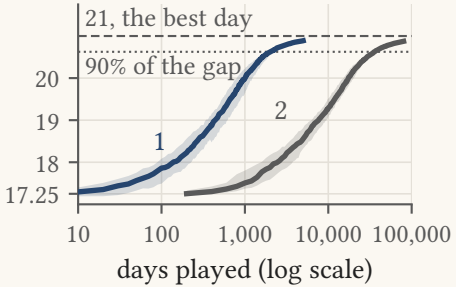


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2 Part 2's method ($N = 100$, $\varepsilon = 0.1$, $\alpha = 0.5$) needs 35,800 days: 200 days played for every move of one cell. Tuned for cost ($N = 5$, $\varepsilon = 1$, $\alpha = 1$) it needs about 900.

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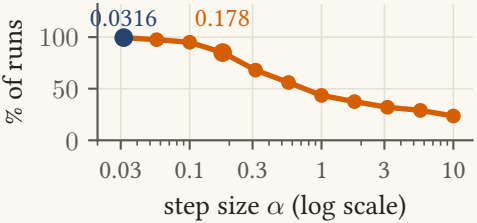
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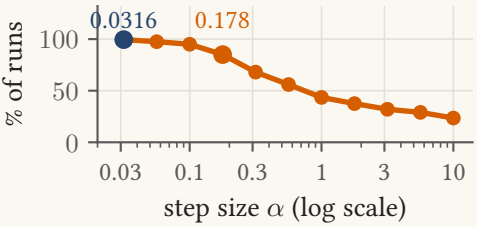
3 One update of REINFORCE moves 13 of the 14 knobs, from 10 days. One update of Part 2's method moves one, from 200 days.

A larger step makes runs fail

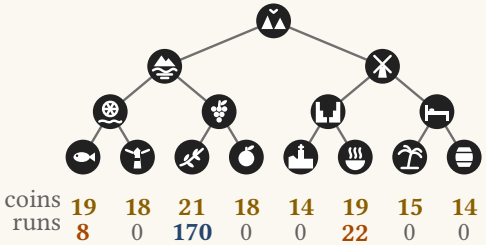


1 At $\alpha = 0.0316$, 199 of 200 runs close 90% of the gap. At $\alpha = 0.178$, 5.6 times larger, 170 do.

A larger step makes runs fail

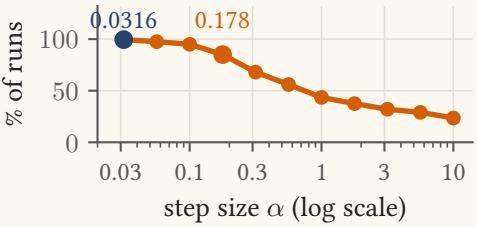


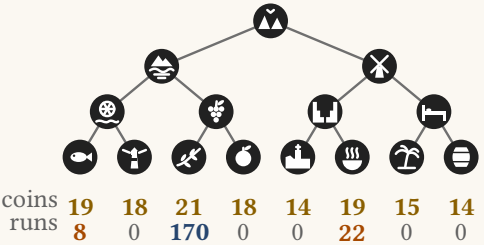
1 At $\alpha = 0.0316$, 199 of 200 runs close 90% of the gap. At $\alpha = 0.178$, 5.6 times larger, 170 do.



2 The other 30 runs end on a day that pays 19 coins. The best day pays 21.

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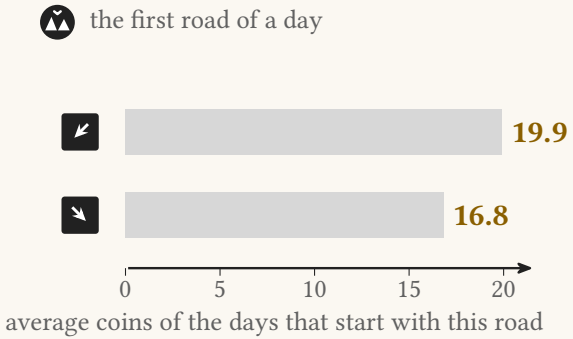


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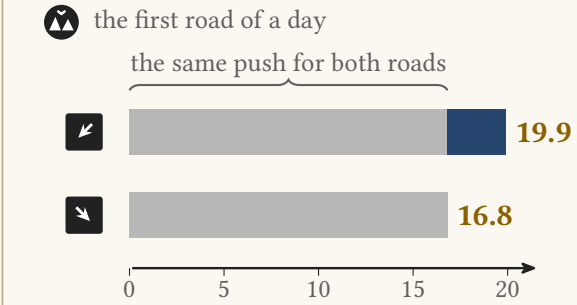
3 Every road taken is pushed up, so with a large step the first roads taken can run away: the runs that fail settle on their day after about 120 days, like the others.

Every road taken goes up, whatever it pays



1 Days that start with ↙ pay 19.9 coins on average, days that start with ↘ 16.8.

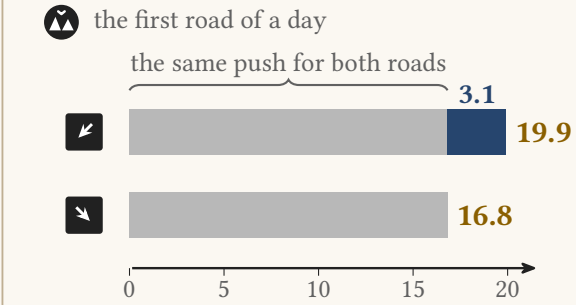
Every road taken goes up, whatever it pays



1 Days that start with ↙ pay 19.9 coins on average, days that start with ↘ 16.8.

2 Each day pushes the road it took by its coins. Coins are never negative, so both roads go up.

Every road taken goes up, whatever it pays



1 Days that start with ↙ pay 19.9 coins on average, days that start with ↘ 16.8.

2 Each day pushes the road it took by its coins. Coins are never negative, so both roads go up.

3 Only the difference, 3.1 coins, tells the better road apart. It is small next to the pushes, and a batch of days can hide it.

A library writes the same update in four lines

```
1 loss = -(G * logp).sum(dim=1).mean()
```

1 `G` and `logp` hold G_t and $\log \pi_w(a_t \mid s_t)$: one row for each of the N days, one column for each of the T steps. The loss sums each row, averages the days and flips the sign.

A library writes the same update in four lines

```
loss = -(G * logp).sum(dim=1).mean()
optimiser.zero_grad()
loss.backward()
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2 `backward` gives the gradient of the loss: minus the average of G_t times the score.

A library writes the same update in four lines

```
loss = -(G * logp).sum(dim=1).mean()
optimiser.zero_grad()
loss.backward()
optimiser.step()
```

1 `G` and `logp` hold G_t and $\log \pi_w(a_t \mid s_t)$: one row for each of the N days, one column for each of the T steps. The loss sums each row, averages the days and flips the sign.

2 `backward` gives the gradient of the loss: minus the average of G_t times the score.

3 A step downhill on the loss is a step uphill on J: the update of the first frame.

Appendix

The algebra behind REINFORCE

Nothing in the frames that follow is needed to follow Part 3.

A.1 the score of a softmax row
A.2 why the average of G_t times the score is the gradient of $\mathcal{J}$

Appendix A.1: the score of a softmax row

A place has a knob for each road, w_1 and w_2, and the chance of road i is $\pi_i = e^{w_i}/(e^{w_1} + e^{w_2})$. ∂_j is the slope in the knob w_j.

$$\log \pi_i = w_i - \log \left(e^{w_1} + e^{w_2} \right)$$

the log of a quotient: top minus bottom

Appendix A.1: the score of a softmax row

A place has a knob for each road, w_1 and w_2, and the chance of road i is $\pi_i = e^{w_i}/(e^{w_1} + e^{w_2})$. ∂_j is the slope in the knob w_j.

$$\log \pi_i = w_i - \log\left(e^{w_1} + e^{w_2}\right)$$

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$$\partial_j \, w_i = \mathbf{1}[j = i]$$

a knob moves only itself

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line 2 minus line 3: the score

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It has mean zero: $\sum_i \pi_i\, \partial_j \log \pi_i = \pi_j - \pi_j(\pi_1 + \pi_2) = 0$, since the chances add up to 1.

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$$\partial_j\, w_i = \mathbf{1}[j = i] \qquad \textit{a knob moves only itself}$$

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Check on the road. Finite differences of $\log \pi_i$ at the 7 places, both roads and seven tables equal $\mathbf{1}[j = i] - \pi_j$ to within 10^{-9}, and the knobs of other places have slope 0. The mean over the road drawn is below 10^{-14}. At the pass of the example the score of the first road is $(+0.29,\ -0.29)$.

Appendix A.2: the gradient of $\mathcal{J}$ as an average

A day d has chance $p_w(d) = \prod_t \pi_w(a_t \mid s_t)$ and average coins $R(d)$, so $\mathcal{J}(w) = \sum_d p_w(d)\, R(d)$. g_t is short for $\nabla_w \log \pi_w(a_t \mid s_t)$. G_t includes the luck.

$$\nabla \mathcal{J}(w) = \sum_d R(d)\, \nabla p_w(d)$$

the product rule; $R(d)$ is fixed by the day

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the product rule; $R(d)$ is fixed by the day

$\nabla p = p\, \nabla \log p$, *the chain rule*

$\log p_w$ *is a sum; the luck averages to 0*

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\nabla \mathcal{J}(w) &= \textstyle\sum_d R(d)\, \nabla p_w(d) && \textit{the product rule; } R(d) \textit{ is fixed by the day} \\
&= \textstyle\sum_d p_w(d)\, R(d)\, \nabla \log p_w(d) && \nabla p = p\, \nabla \log p \textit{, the chain rule} \\
&= \mathbb{E}\Big[\, G_1 \textstyle\sum_t g_t \Big] && \log p_w \textit{ is a sum; the luck averages to 0} \\
&= \mathbb{E}\Big[\, \textstyle\sum_t \boxed{G_t}\, g_t \Big] && \textit{earlier coins meet a mean-zero score (A.1)}
\end{aligned}
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\nabla \mathcal{J}(w) &= \textstyle\sum_d R(d)\, \nabla p_w(d) && \textit{the product rule; } R(d) \textit{ is fixed by the day} \\
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\end{aligned}
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Check on the road. At seven tables lines 1 and 2 equal the library's exact gradient to within 10^{-8} and lines 3 and 4 to within 10^{-12} (exact averages over the 8 paths and their 27 luck combinations). $\mathbb{E}[r_k\, g_t]$ is below 10^{-14} for $k < t$ and 0.82 for $k = 2$, $t = 1$ at the example knobs.